

Laxminarayana Moru
Certified Azure Data Engineer

Mobile: (945) 360-8282

Email: laxman.moru@gmail.com

Professional Summary:

With 13+ years of experience in providing data solutions—including design, architecture, and development—to clients such as Microsoft, A.J.G., Hyperion Insurance Brokerage, Amazon, and Providence Healthcare. I have extensive expertise in **Azure Databricks, Apache Spark, Python, Pyspark, Spark SQL, Data Warehousing, Data Modeling, Airflow, MS SQL Server, and various Azure services like Azure Synapse, Azure Data Storage, Azure Data Factory, Azure Databricks, Azure SQL, and SSIS.**

- **Certified Azure Data Engineer (DP-203)** with 13+ years of experience in data world
- Extensive experience in designing, developing, and deploying end-to-end ETL/ELT solutions using **Azure Databricks, Apache Spark, Python, Pyspark, Spark SQL, Lakehouse architecture, Azure Synapse, Azure Storage, Azure Data Factory, Airflow, SQL Server Integration Services (SSIS), and SQL Server.**
- Skilled in loading raw data from heterogeneous sources into the Enterprise Data Warehouse (EDW) using **Azure Data Factory and SSIS.**
- Proficient in building dynamic **Azure Data Factory** pipelines through parameterization of Linked Services, Datasets, and dynamic auto-mapping of source and sink.
- Experienced in loading data via Azure Data Factory from various sources, including **SQL Server, MySQL, JSON, XML, CSV, SFTP, Salesforce, Snowflake, and HTTP.**
- Knowledgeable in utilizing **Azure Blob Storage and Azure Data Lake Store.**
- Hands-on experience with **Azure Databricks Notebooks, using Pyspark and Spark SQL** to process and transform large datasets.
- Designed and implemented Medallion architecture with **Delta Lake in Azure Databricks, utilizing Auto Loader, Unity Catalog, and Delta Live Tables.**
- Expertise in **Azure Synapse Analytics**, a massively parallel processing data warehouse, including data loading via **PolyBase.**
- Proficient in using **Azure Key Vault** to securely store connection strings and secrets.
- Experienced in building data pipelines using **Apache Airflow**
- Expertise in designing **OLTP and OLAP data models.**
- Implemented a **knowledge graph with Neo4j** by processing unstructured data using open-source LLMs such as Mistral, leveraging prompt templates.
- Developed vector embeddings with **Mongo Atlas** collections to enable semantic search, employing **Retrieval-Augmented Generation (RAG)** for custom LLMs.
- Strong technical expertise in **MS SQL Server**, including performance tuning, reporting, and troubleshooting.
- Thorough knowledge of addressing performance issues, including query tuning.
- Experienced with Control Flow, Data Flow, Event Handlers, and Logging for SSIS packages.
- Skilled in monitoring production SSIS packages and troubleshooting issues or failures.

- Knowledge and proficiency in utilizing Git repositories for effective code base management, version control, and the ability to mentor and support peers.
- Collaborated with end users and customers to understand business requirements, recommend technical solutions, and document functional requirements.

Work Experience:

- Worked as **Principal Data Engineer** in Providence Global Centre, Hyderabad from January 2023 to January 2025.
- Worked as **Data Engineer** in **Amazon**, Hyderabad from September 2022 to December 2022.
- Worked as a **Lead Data Engineer** in **Accenture (Earlier known as Pramati Technologies)**, Hyderabad from Jan 2014 to Aug 2022.
- Worked as a **Software Engineer** in **MAQ Software**, Hyderabad from Sept 2011 to Jan 2014.

Educational Qualification:

- Master of Computer Applications (MCA) from Osmania University with 78% (2008-11).
- Bachelor of Science from Kakatiya University with 83% (2005-08).
- Board of Intermediate Education with 85% (2003-05). • SSC from K.H.S with 82% (2002-03).

Technical Skills:

- Azure Data Factory, and Azure SQL Database
- Azure Blob & Data Lake Store
- Azure Databricks, Pyspark, SparkSQL, Python
- Azure Synapse Analytics, Snowflake, and Airflow
- Gen-AI features like Vector Embedding (Mango Atlas) and Semantic Model (Knowledge Graph using Neo4j)
- Microsoft SQL Server
- Microsoft SQL Server Integration Services

Project #10:

Dec 2023 to Jan 2025

Project Title: RMS (Registry Management System)

Role: Principal Data Engineer

Environment: Azure SQL, Python, Neo4j, Mongo Atlas Vector database, Airflow, and Snowflake

Description:

The Registry Management System (RMS) deals with various disease registries such as Strokes, Heart Failure, Cardiovascular Disease (CVD), and Neonatal Intensive Care Units (NICU). In the USA, Healthcare facilities are required to submit patient health and treatment details to different regulatory bodies, such as GWTG, NCDR, and VON. We typically consume data from the EPIC EHR system and process it by loading it into UDM and registry-specific data models, both discrete and non-discrete, for review by an abstractor. The abstractor then submits the data to the regulatory body upon completion of the review process. Discrete fields come directly from the Clarity database, an intermediary layer between EPIC EHR and our system, while nondiscrete fields are extracted from patient notes using open-source large language models (LLMs) like Mistral, employing prompt templates.

Responsibilities:

- Analyze EPIC EHR data to identify the required fields for each registry.
- Implement a data engineering pipeline to process the identified fields from EPIC EHR and load them into UDM and registry-specific data models.
- Process patient notes (unstructured text data) and extract fields using open-source LLMs like Mistral with prompt templates, then load the extracted data into the Neo4j database to construct a knowledge graph.
- Perform data accuracy checks against ground truth data once development is complete for each registry.
- Lead the data engineering team in both technical and functional activities, including development and design.
- Participate in all agile sprint ceremonies.
- Handle various types of production issues.

Project #10:**Jan 2023 to Nov 2023****Project Title: PEAP (Patient Experience Analytical Platform)****Role:** Principal Data Engineer**Environment:** Python, Pyspark, ADO, ADF, snowflake, Apache Spark, Azure Data Lake**Description:**

PEAP is a patient experience system which is designed to capture feedback and insights about the inpatients and extract derivatives out of it to form mandatory questionnaire governed by US healthcare and then recapture the answers from the patients and run analytics on top of the obtained results.

Responsibilities:

- Analyze the requirements and design the architecture.
- Involved in consuming data from various upstream viz snowflake.
- Writing various kind of aggregations and transformations according to business needs using Spark SQL and Spark Data Frame functions.
- Analyzing and coding new pipelines using python and pyspark to replace existing ADF pipelines to make the product resilient and robust.

- Involved in handling different types of production issues.
- Involved in performance tuning of spark applications.
- Leads the Data-Engineering development team in terms of technical and functional ways.
- Involved in all agile-sprint ceremonies.

Project #9:

Sep 2022 to Dec 2022

Project Title: Automated Inventory

Management

Role: Data Engineer

Environment: Spark, RedShift, S3, SQL, Cradle&Datanet (Internal tools), Quicksights, Python

Description:

Automated Inventory Management is part of SCOT (Supply Chain Optimization Technology) and is designed to identify the defects across the SCOT four verticals (Forecasting, Sourcing, IPC and FO) and fix them in automated way to improve the Amazon sales across the WW.

Responsibilities:

- Analyze the requirements and design pipelines
- Build the data pipelines to consume the upstream data and process them into Business Metrics (OOS, FT, GVS, FOOS etc..)
- Provide proactive On-call support activities

Project #8:

Feb 2022 to Aug 2022

Project Title: Complete Source Migration

Client: Howden Insurance Group, UK **Role:** Lead

Data Engineer

Environment: Azure Data factory, Azure Data Lake, PySpark with Databricks, and Azure SQL Datawarehouse.

Description:

As part of Partial Migration, we migrated the semi processed data from Accelerator 1.0 Staging into new Data Platform to meet the Business timelines to deliver key dashboards against the new Data Platform. Now, we need to switch to ingesting the raw data from source system itself and process it in Data Platform itself rather than semi processed data from Accelerator 1.0 Staging.

Responsibilities:

- Involved in design of source ingestions including POCs
- Leading the team of 5 developers on technical front including design with regular hands on with development including ADF, Data Lake, and PySpark with Databricks.

- Conduct the reviews to ensure the implementation following business requirements along with proposed design and coding standards

Project #7:

August 2021 to Jan 2022

Project Title: BU Re-design

Client: Howden Insurance Group, UK

Role: Azure Data Engineer and Technical Lead

Environment: Azure Data factory, Azure SQL Server and SQL Server 2019 BDC

Description:

Business Unit is a key classification of HIG Business, Customer more interested to analyze the data in Dashboards/Reports from Business Unit point of view. However, legacy design of Business Unit in Accelerator is become an over-headache to Business to handle certain business scenarios like business split into two, merging multiple business unit into one etc.... and ETL team is also suffering to manage every FY roll-over work for Business Unit due to the legacy design and taking lot of manual effort for every FY Roll-over. Hence Business decided to re-design the Business Unit module to make job easy for everyone including Data Stewards, ETL team, and Business team.

Responsibilities:

- Involved in end-to-end design of Business Unit Re-design project
- Contributed with key inputs in designing the new UI application to manage BU work
- Implemented generic ADF Pipeline with scalable meta-data configuration to refresh the BU information in EDW

Project #6:

Dec 2020 to August 2021

Project Title: DUAL

Client: Howden Insurance Group, UK

Role: Azure Data Engineer and Technical Lead

Environment: Azure Data factory, Azure SQL Server, Azure Blob & Data Lake store, PySpark with Databricks, and Azure SQL Datawarehouse

Description:

DUAL is an underwriting division of the Howden Insurance Group. It is the world's largest international underwriting agency and Lloyd's of London's largest international cover holder. Built dynamic pipelines using Azure Data Factory to load underwritings data from on premise SQL server source → pre-staging → Azure Blob Storage → Azure Data Lake → Staging layer → enterprise Data Warehouse (EDW). The data loaded into EDW is used for reporting in Dashboards.

Responsibilities:

- Understanding the business requirement and mapping document provided by the modeling team.
- Built generic pipelines using Azure Data Factory to extract data from SQL Server.

- Built dynamic ADF pipelines by storing the automapping source to sink json in an azure sql server configuration table.
- Used Azure Blob Storage and Azure Data Lake Store.
- Implemented Transformations according to the Business rules using PySpark with Databricks

Project #5:

Jan 2020 to Dec 2020

Project Title: Partial Migration Client:

Howden Insurance Group, UK

Role: Azure Data Engineer and Technical Lead

Environment: ADF, Azure Storage, PySpark, Azure SQL, SQL Server 2019 BDC.

Description:

Partial Migration is about migrating the existing data from On-premises system to brand new Data Platform which was built with cloud technologies as mentioned below. However, migration scope of this project is partial due to the foreseen business requirements for key dashboards/reports to be delivered against the new Data Platform for migrated data. Hence scope of the project was limited to semi processed data from existing system but not complete migration including source ingestion. **Responsibilities:**

- Involved in strategy and design of Partial Migration project
- Enhanced the existing ADF Pipeline to make them more generic
- Enhanced existing PySpark with Azure Databricks Notebooks design/implementation to extend its functionality to cater more in addition to Batch validation, Delta Calculation, and Data Validation.
- Designed and built generic Rule engine to refresh the EDW with high reusability and more

Project #4:

July 2018 to Jan 2020

Project Title: Data Platform Client:

Howden Insurance Group, UK

Role: Azure Data Engineer and Technical Lead

Environment: ADF, Azure Storage, PySpark, SQL Server 2019 BDC.

Description:

Data Platform is nothing but cloud version of Accelerator which was built on legacy technologies like MSBI. As part of this project, we designed and implemented the framework using complete Azure cloud tech stack ADF, Azure Storage (BLOB and Lake Gen2), PySpark, and SQL Server 2019 BDC. This is a Cloud framework which includes generic ADF pipelines including error handling and restart-ability, Custom Logging, Spark which handles Delta identification and Data validation for any future ingestion by configuring some basic metadata.

Responsibilities:

- Involved in end-to-end design of Data Platform including generic ADF Pipelines, PySpark, Custom Logging.
- Implemented generic ADF Pipeline to ingest the data

- Involved in PySpark development with Azure Databricks Notebooks to implement the Batch validation, Delta Identification and Data validation modules
- Have good hands-on Azure BLOB and Azure Data Lake Gen2

Project #3:

Apr 2017 to June 2018

Project Title: Accelerator

Client: Howden Insurance Group, UK

Role: Azure Data Engineer and Technical Lead

Environment: SQL Server 2012, SSIS 2012, Azure Analysis Services.

Description:

Howden Insurance Group is one of the largest Insurance Brokerage companies in the UK and has subsidiaries across the world. HIG operates mainly in Insurance Brokerage, Re-Insurance Brokerage, and Underwriting divisions. The Business Intelligence Data warehouse (BI DW) involves development, maintenance, and monitoring of reports and other data warehouse related activities.

The main goal of this project is integrating the data from all subsidiary companies of HIG across the world into a warehouse called Accelerator to provide more insights for Business through building various reports called QlikView, Power BI, and Angular on top of Azure Analysis Services.

Responsibilities:

- Understanding the business requirement and communicating with business users as required when developing this model.
- Involved in design for any new requirements/feature implementation requested by Business
- Used to lead the team in technical and design perspective for any new feature integration or any challenge faced by the team members
- Understood the business process of HIG in quick time and used the deliver the work with ease as business requirements more tightly coupled with customer business process
- Involved in developing SSIS packages and Stored Procedures to load the data into a data warehouse by using SSIS.
- Ingested new sources called (Howden India, Howden Dubai, Howden Sweden, Howden Norway) into the existing Accelerator Datawarehouse by doing gap analysis on my own, building the mappings between new source system to existing Accelerator Warehouse.

Project #2:

Jan 2014 to Mar 2017

Project Title: SMART

Client: AJG Gallagher, USA

Role: ETL and Report Developer

Environment: SQL Server 2012, SSIS 2012, SSAS 2012

Description:

As part of this project, we delivered various OLTP applications called IDC, LR, LSS, ProGen, TimeSheet etc... SMARTs applications will enable the team to process and track the requests through different stages in the IDC process. The goal of IDC is to integrate with DMS and EPIC applications and provide GSC teams with a seamless experience, thus eliminating redundancy and improving efficiency in the associated stages of the process. It will also enable in defining the teams based on skill sets and improve the work assignment process.

Responsibilities:

- Collected business requirements from users and translated them as technical specs specifications.
- Involved in OLTP database design as per the business requirements.
- Implemented Stored Procedures to handle DML operations, Search queries, get queries w.r.t the action taken from Application
- Optimized backend queries/stored procedures to achieve the quick response time
- Involved in deploying SSIS packages in Integration services Catalogs and created Jobs.
- Involved in developing SSIS packages to load the data from the OLTP database to Reporting database.
- Handled ETL failures/issues.
- Created Pivot reports on top of Reporting database

Project #1:**Oct 2011 to Jan 2014****Project Title: RIO Pipeline Client:**

Microsoft, USA

Role: ETL and Report Developer

Environment: SQL Server 2008/R2, SSIS 2008/R2, SSAS 2008, Java Script, .Net.

Description:

The main goal of the project is to track how Prospect can be converted into an Opportunity for Microsoft Products with various marketing campaign strategies called Inbound and Outbound. This model contains the data about the Prospect, Activity, Contacts, Markets, Opportunities etc... We used to have a multi-dimensional SSAS cube on top of the Warehouse to feed the various SSRS, HTML-5, and Adhoc reports.

Responsibilities:

- Involved in developing packages to load the data into a data warehouse by using SSIS containers, tasks and transformations.
- Created Stored Procedures, views and other SQL statements for SSIS package and Warehouse data processing.
- Involved in developing the multi-dimensional cube for main and Adhoc reports
- Optimized SSIS packages and stored procedures.
- Handled different types of production ETL failures/Issues.